

Project 1
Two-Stage Miller-Compensated OTA with single-ended output and differential-output

Due Date: 11/30/2018

Midnight 11:59 PM

Total points: 100

For this project, you will design a two-stage differential input single-ended output CMOS operational amplifier that meets or exceeds the following specifications.:

- Load $\geq 1\text{pF}$
- Technology = 130nm CMOS
- $V_{dd} \leq 1.3\text{V}$
- Low-frequency voltage gain $> 60\text{dB}$
- Low-frequency common mode rejection ratio $> 80\text{dB}$ (introduce **3-sigma offset**.
1-sigma offset = $5\text{mV}/\sqrt{WL}$) where W and L are the sizes of your input differential pair for the first-stage (in micron). For eg, if W and L are 10u and 1u each, the 1-sigma offset is 1.58mV.
- Unity-gain bandwidth $> 800\text{MHz}$
- Phase margin $> 52^\circ$
- Output swing: 0.3Vpk-pk-se
- Input common-mode range: $0.5\text{V} < V_{ic} < 0.8\text{V}$
- Large-signal slew-rate $> 50\text{V}/\mu\text{sec}$ for two cases:
 - Unity-gain (voltage follower config.) with 0.9V to 0.4V voltage step applied to input
 - Unity-gain (voltage follower config.) with 0.4V to 0.9V voltage step applied to input
- Power consumption and Input-referred Noise: minimize
- Architecture: Your choice (Either PMOS or NMOS or PMOS+NMOS input stage is ok). Please use telescopic cascode amplifier for the first-stage and a common-source amplifier (can be cascode) for stage 2.
- The Phase 1 incorporates the 2-stage amplifier in open-loop.

476 students and MS students (Non-Microelectronics Track):

Please submit **one report**. The report should include

- First-pass design calculations for all bias conditions, transistor sizes, small-signal performances, schematics, etc. (for both designs) (25 points)
- Clearly label Cadence schematic and simulation results showing that all specifications are met for both designs. (50 points)
- Report presentation (25 points)

MS Thesis/Non-Thesis (Microelectronics):

- First-pass design calculations for all bias conditions, transistor sizes, small-signal performances, schematics, etc. (for both designs) (25 points)
- Clearly label Cadence schematic and simulation results showing that all specifications are met for both designs. Your simulation need to include bias circuits for generating any biases that you have used (50 points).
- 4-page report in ISCAS format (Link: <http://tcas1.polito.it/general.html>).

The 2 attached papers are for reference for the closed-loop analysis in Phase-II design using these specifications.